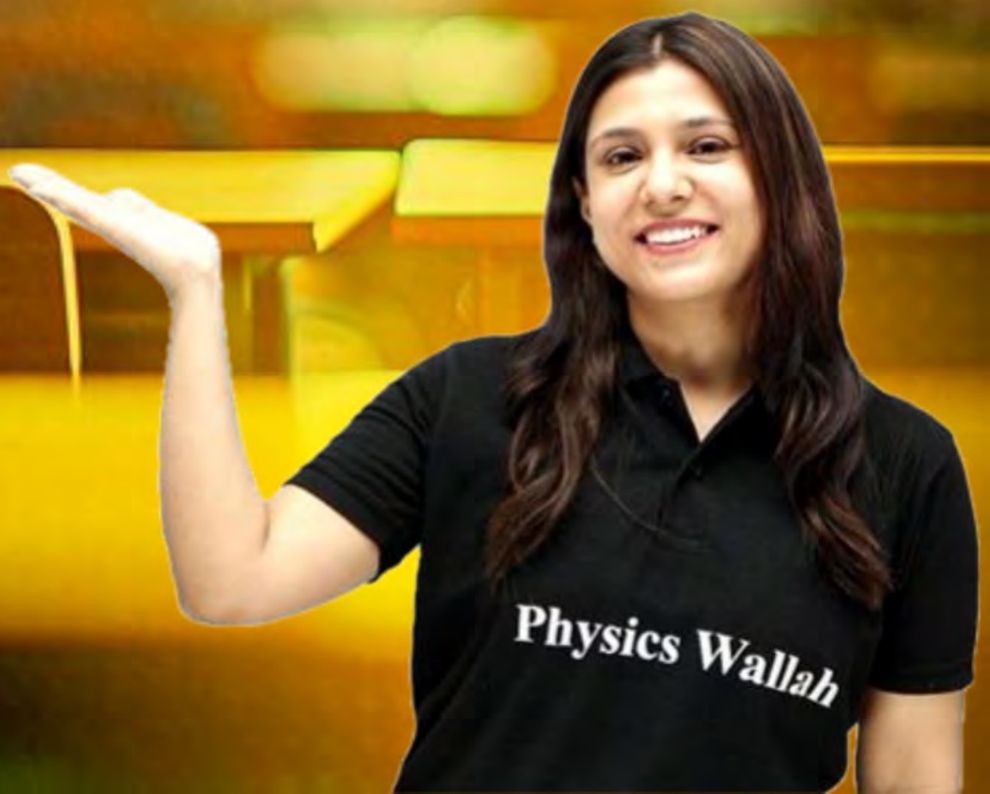
RADIANT 2026

Biology

Digestive System

Lecture - 02

By - Upasana Mongia Ma'am



Topics to be covered

1. Digestion in Mouth
2. ——— Stomach.
3. Glands and their functions-1
4.



Glands

Glands are special areas in our body [cell, tissue, organ] which secretes some chemicals.

For. E.g → Salivary Glands.

↓

Saliva [Juice]

↓

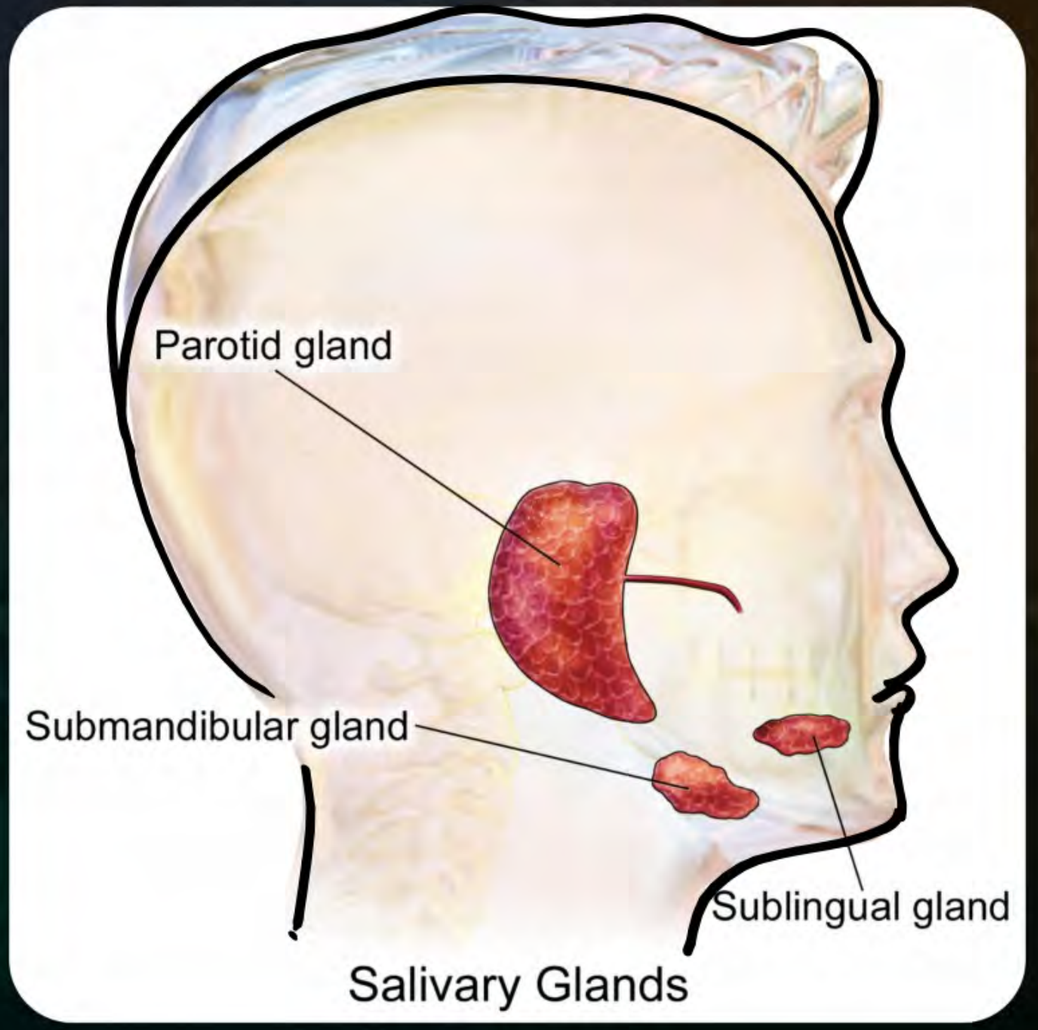
Salivary amylase [Enzymes]



Salivary Glands

- There are three pairs of salivary glands in the wall of the buccal cavity which opens into the mouth They are the parotid glands, submaxillary glands and sublingual glands.

- Salivary glands secrete a watery fluid called saliva, which initiates digestion in the buccal cavity



Salivary Glands

Digestion of ——— Starts in the mouth.

A) Proteins

B) Fats

C) Carbohydrates

Functions of Saliva

- Salivary amylase present in the saliva converts the starch into maltose [redacted]
- The mucus present in the saliva lubricates the food and thus helps in the swallowing of food.
- Saliva moistens and lubricates the inner lining of the mouth cavity and the surface of the tongue and helps us in speaking.

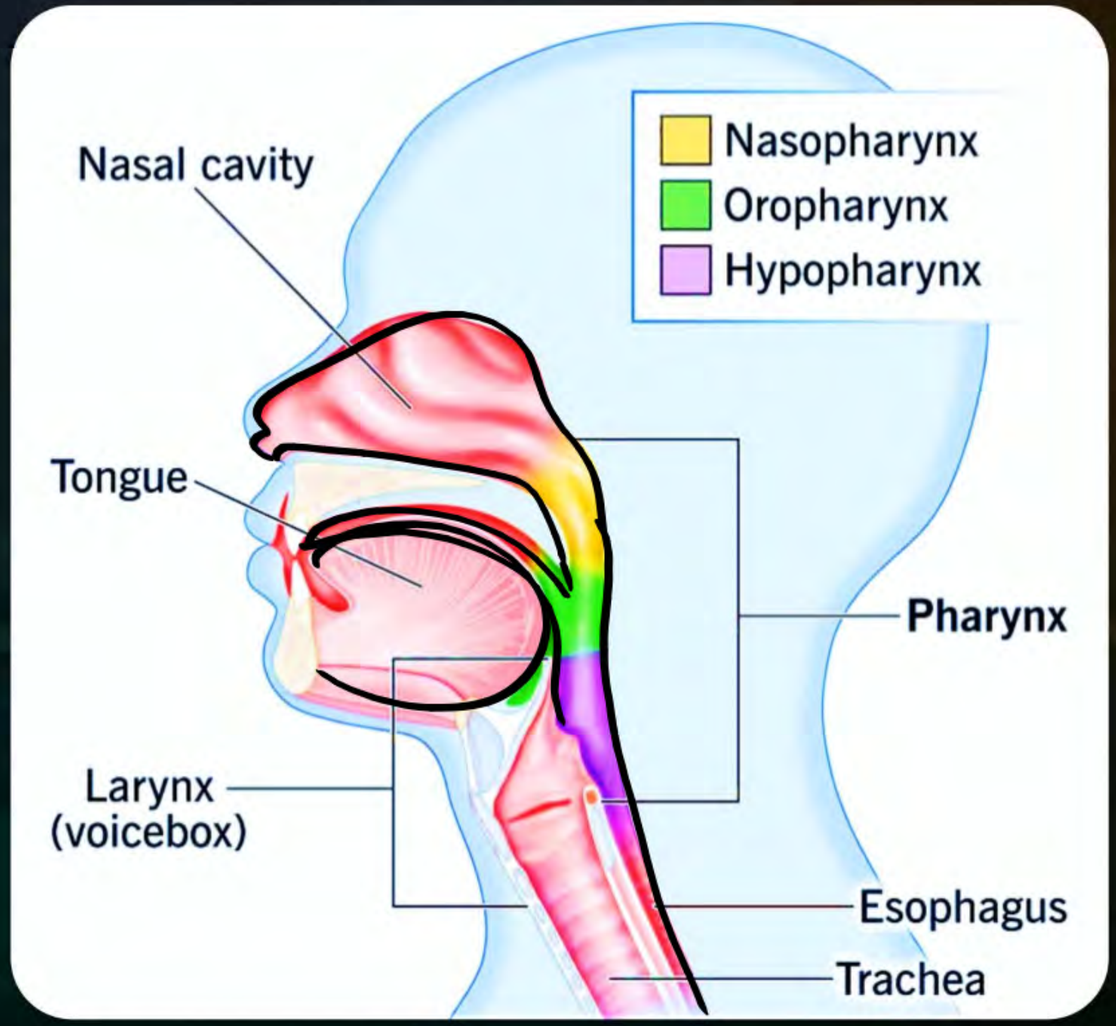
Aloo paratha



Carbohydrates (30%) —Salivary amylase→ Maltose (Glucose + Glucose)

Pharynx

- The pharynx is the common passage for food and air leading to the oesophagus and the larynx, respectively.

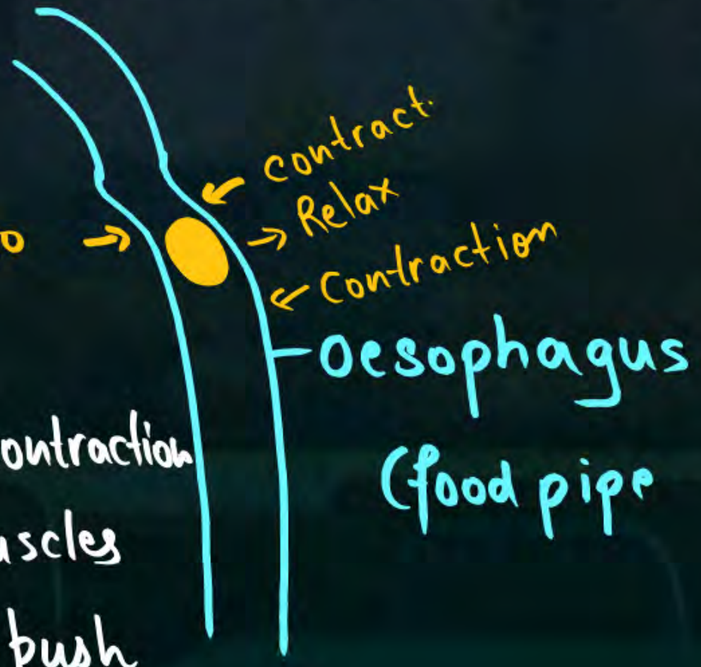


Functions of the Pharynx

- Food passes through the pharynx to the oesophagus.

Oesophagus

- Connects Mouth to Stomach.
- Peristalsis → The contraction and relaxation of Muscles of Oesophagus to push the food down.



Peristaltic Movement:

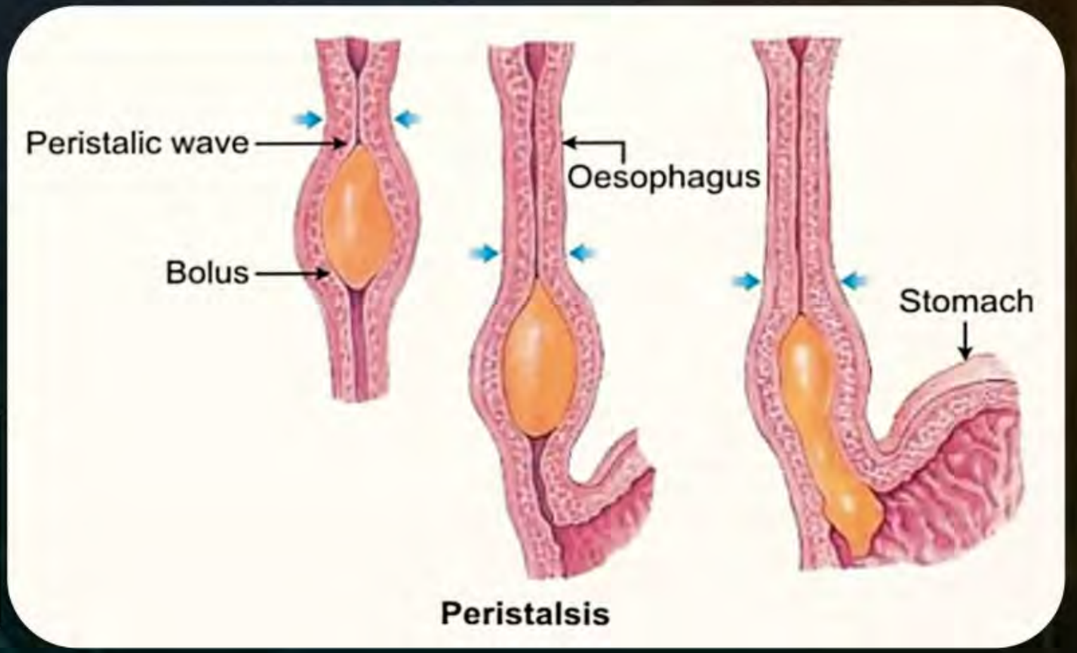
Functions of the Oesophagus

- [redacted]

- The oesophagus has a sphincter at the front end of the stomach to prevent the backflow of food from the stomach.

- Swallowing is the process which involves the movement of food from the mouth to the stomach through the oesophagus.
- Swallowing involves three phases-buccal phase, pharyngeal phase and oesophageal phase.

- Peristalsis is the rythemic contraction and relaxation of muscles of the oesophagus.

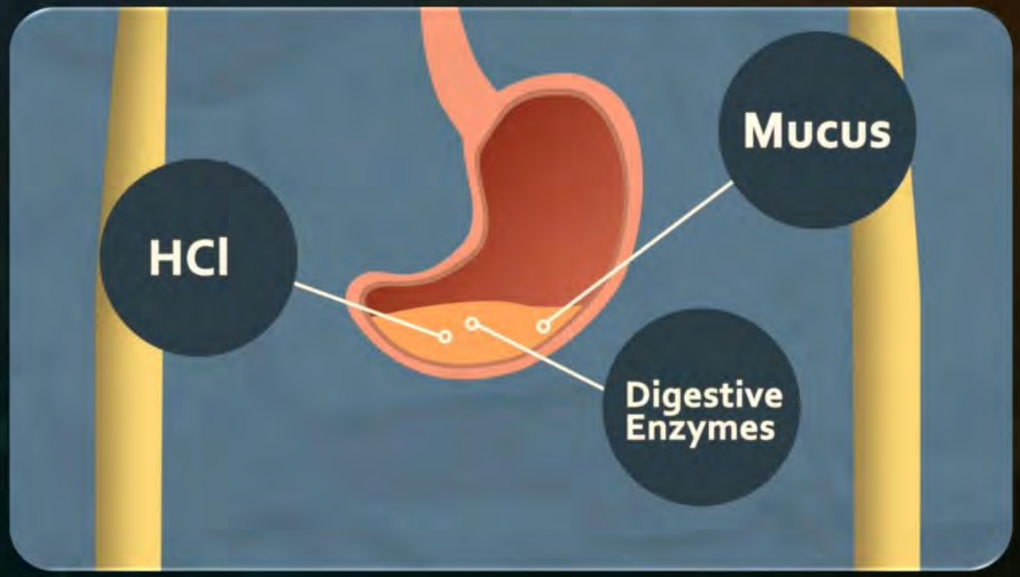


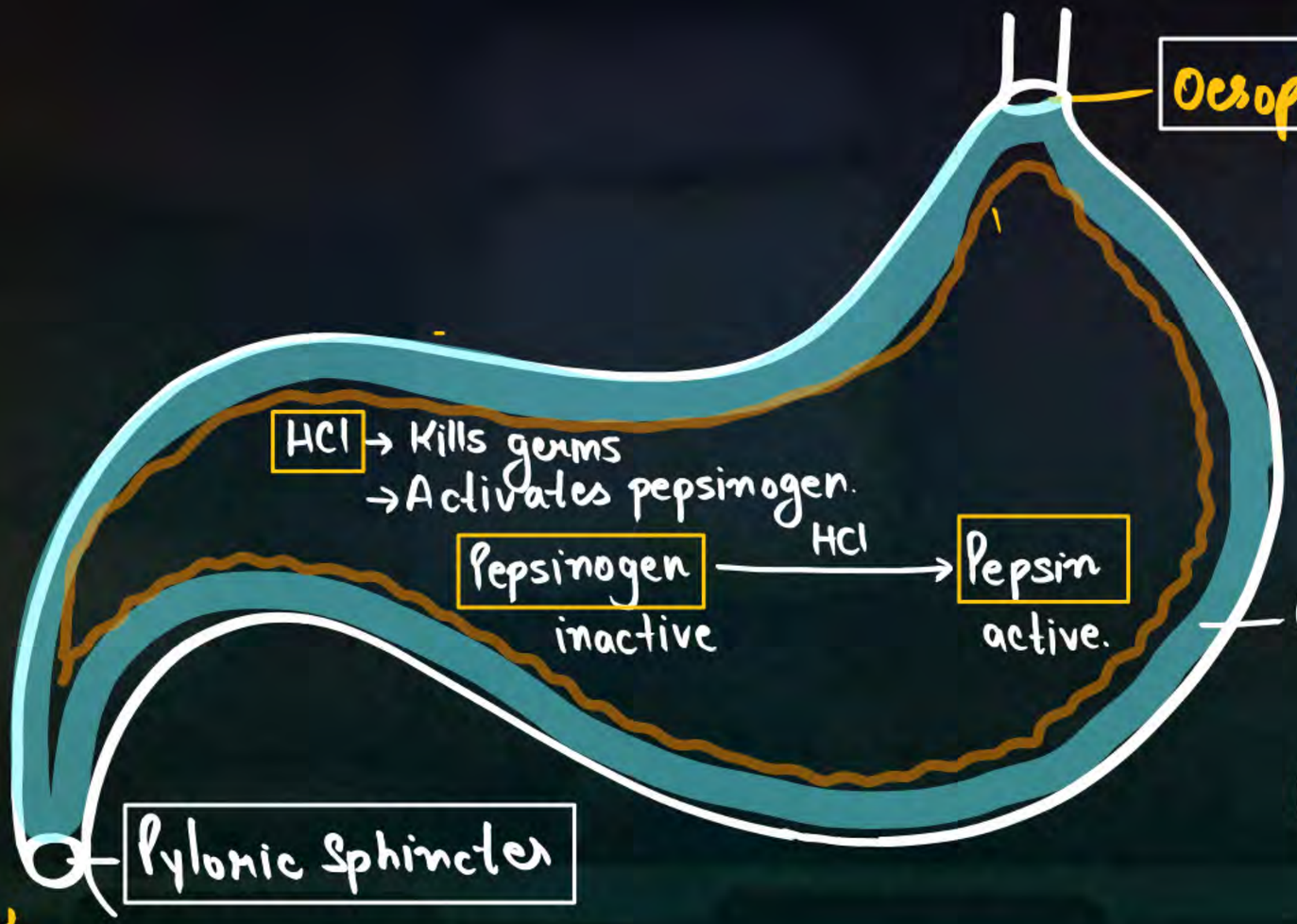
Peristalsis

Stomach

- The stomach is a J-shaped muscular, bag-like organ with walls made of thick and elastic muscles.

- The gastric glands release three gastric juices-hydrochloric acid, enzyme pepsin and mucus.





Oesophageal Sphincter.

- opens into stomach
- prevents backflow of food

Chyme → acidic food in stomach.

HCl → Kills germs
→ Activates pepsinogen.

Pepsinogen (inactive) —HCl→ Pepsin (active).

Gastric Glands → Gastric Juice → HCl, Pepsinogen (inactive), Mucus

Pylonic Sphincter — opens into duodenum of Small Intestine

Infants

Prorenin —HCl→ Renin.

Caesin —Renin→ Para-Caesin

Proteins —Pepsin.→ peptides + peptones

The digestion of fats. start in

A) Mouth

B) Stomach

C) Oesophagus

D) None of the above.


Thank You